FABLE-Pakistan: A Data Pipeline for Pakistani Agricultural Production and Trade

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Abstract

FABLE (Food, Agriculture, Biodiversity, Land-use and Energy) is an Excel-based tool, developed by IIASA and SDSN, that models a country's food and land-use system out to 2050. For Pakistan it currently runs on generic FAOSTAT figures, which do not always match what Pakistan's own government publications report. This project builds a Pakistan-specific alternative: a provincial crop and livestock production dataset drawn from several years of official statistics, a national trade dataset pulled from UN Comtrade and mapped through a tariff-code concordance table, and a validation pass comparing the result against FAOSTAT itself. Building this pipeline surfaced a number of real data-quality issues along the way, including tariff-code reclassifications that had silently erased years of trade in certain commodities, a byproduct code that had inflated an official export figure, and a handful of transcription errors sitting in the original source publications. Each of these was traced to its root cause and corrected where possible. This report explains what was built, how it was built, and what the resulting figures show; the full technical detail behind every individual decision is documented separately for reference.

Contents

1 Introduction

FABLE runs as ten linked calculation steps inside a single spreadsheet — consumption, livestock production, crop production, land use, and finally trade and food-security indicators — and it was built to work for any country using the same generic FAOSTAT inputs. Pakistan, however, does publish its own detailed agricultural statistics every year, at the provincial level, and those numbers do not always agree with what FAOSTAT reports for the same crop and year. This project's job was to close that gap: replace FABLE's Pakistan inputs with data pulled directly from Pakistan's own sources, and check how the result compares to FAOSTAT so the difference is understood rather than just assumed away.

The work was completed over four weeks as part of the 2026 Summer Mentorship Program at LUMS. It moved through several connected stages: matching FABLE's own commodity list against what Pakistan actually produces and trades, building a provincial production dataset, building a national trade dataset through a tariff-code concordance table, joining the two into a single balance table, and validating the result against FAOSTAT. Each stage is covered in turn below.

2 Data Sources

Table 1 lists the main sources used. Production figures come almost entirely from Pakistan's own agricultural and horticultural statistics, and trade figures come from UN Comtrade. FAOSTAT is used only as the baseline the trade dataset is checked against at the end of the pipeline, never as a source for filling gaps earlier in it.

Category	**Sources**
Crop production	MINFAL Agricultural Statistics of Pakistan (ASP), 2021-22 and 2023-24 editions; PBS Agricultural Statistics of Pakistan 2010-11; PBS District-Wise Crops, 2022-23 and 2024-25 editions.
Fruit, vegetable and condiment production	Fruit, Vegetables and Condiments Statistics of Pakistan (FVC), five editions spanning 2008-09 through 2023-24.
Livestock production	ASP Tables 122–125 (milk, meat and eggs, other livestock products, fish), 2021-22 and 2023-24 editions.
Trade flows	UN Comtrade, queried with Pakistan as reporter.
Conversion factors	FAO Technical Conversion Factors (Pakistan); FAO Food Balance Sheets Handbook; ICCO cocoa statistics; IRRI rice-milling data; USDA Agricultural Handbook 697; FAO/JECFA's guar gum assessment; industry references for smaller commodities.
Validation baseline	FAOSTAT, Trade: Crops and Livestock Products domain.

Table 1: Main data sources used in the pipeline.

3 FABLE's 76 Commodities

FABLE classifies the global food system into 76 individual commodities, grouped into 19 broader categories such as cereals, oilseeds, pulses, and livestock products. Before any data could be collected, each of these 76 had to be checked against what Pakistan actually produces and matched to the Pakistani publication, or combination of publications, that could supply it. Table 2 lists all 76, grouped the way FABLE groups them, together with how each was ultimately classified: whether direct provincial data exists in an official Pakistani publication, whether the figure has to be built by summing several Pakistani crops together, whether it can only be reached by applying a conversion factor to a related raw material, or whether Pakistan does not produce or trade it at all.

This classification was the starting point for the production dataset, not a guarantee. A few commodities initially expected to have usable data turned out, once the actual publications were checked line by line, to have no separate series after all; these cases are noted again in the limitations at the end of this report.

#	Commodity	Group	Classification
1	Wheat	Cereals	Available
2	Rice (paddy)	Cereals	Available
3	Corn / Maize	Cereals	Available
4	Millet	Cereals	Available
5	Sorghum	Cereals	Available
6	Barley	Cereals	Available
7	Oats	Cereals	Not applicable
8	Rye	Cereals	Not applicable
9	Other cereals	Cereals	Not applicable
10	Sugar cane	Sugar	Available
11	Sugarbeet	Sugar	Available
12	Sugar raw	Sugar	Needs conversion factor
13	Cotton lint	Fibre & Industrial	Available
14	Jute	Fibre & Industrial	Available
15	Rubber	Fibre & Industrial	Not applicable
16	Other hard fibers	Fibre & Industrial	Not applicable
17	Other soft fibers	Fibre & Industrial	Not applicable
18	Cotton (seed)	Oilseeds	Available
19	Rapeseed	Oilseeds	Available
20	Sesame	Oilseeds	Available
21	Sunflower	Oilseeds	Available
22	Oil palm fruit	Oilseeds	Not applicable
23	Olive	Oilseeds	Not applicable
24	Soybean oil	Oilseeds	No local data
25	Coconut oil	Oilseeds	Needs conversion factor
26	Groundnut oil	Oilseeds	Needs conversion factor
27	Palm oil	Oilseeds	Not applicable
28	Other oils	Oilseeds	Built by summing crops
29	Cotton cake	Oilseed Cake	Needs conversion factor
30	Groundnut cake	Oilseed Cake	Needs conversion factor
31	Rapeseed cake	Oilseed Cake	Needs conversion factor

(continued)

#	Commodity	Group	Classification
32	Other oilseed cake	Oilseed Cake	Needs conversion factor
33	Gram / Chickpea	Pulses	Available
34	Mung bean	Pulses	Available
35	Lentil / Beans	Pulses	Available
36	Peas	Pulses	Available
37	Other pulses	Pulses	Built by summing crops
38	Potato	Roots & Tubers	Available
39	Sweet potato	Roots & Tubers	Available
40	Cassava	Roots & Tubers	Not applicable
41	Yams	Roots & Tubers	Not applicable
42	Other tubers	Roots & Tubers	Not applicable
43	Apple	Fruit & Vegetables	Available
44	Banana	Fruit & Vegetables	Available
45	Coconut	Fruit & Vegetables	Available
46	Date	Fruit & Vegetables	Available
47	Grape	Fruit & Vegetables	Available
48	Grapefruit	Fruit & Vegetables	No local data
49	Lemon	Fruit & Vegetables	No local data
50	Onion	Fruit & Vegetables	Available
51	Orange (Citrus)	Fruit & Vegetables	Available
52	Pineapple	Fruit & Vegetables	Not applicable
53	Plantain	Fruit & Vegetables	Not applicable
54	Tomato	Fruit & Vegetables	Available
55	Other citrus	Fruit & Vegetables	Built by summing crops
56	Other fruits	Fruit & Vegetables	Built by summing crops
57	Other vegetables	Fruit & Vegetables	Built by summing crops
58	Nuts	Nuts	Available
59	Tobacco	Beverages & Spices	Available
60	Tea	Beverages & Spices	Not applicable
61	Cocoa	Beverages & Spices	Not applicable
62	Other spices	Beverages & Spices	Built by summing crops
63	Fish	Fish	Available
64	Butter / Ghee	Animal Fat	Available
65	Cream	Animal Fat	No local data
66	Raw animal fat	Animal Fat	Available
67	Eggs	Eggs	Available
68	Milk	Milk	Available
69	Poultry meat	Poultry	Available
70	Beef	Red Meat	Available
71	Goat & Lamb	Red Meat	Available
72	Pork	Pork	Not applicable
73	Offal	Other	No local data
74	Beer	Alcoholic Beverages	Not applicable

(continued)

#	Commodity	Group	Classification
75	Wine	Alcoholic Beverages	Not applicable
76	Other alcoholic bev.	Alcoholic Beverages	Not applicable

Table 2: FABLE's 76 commodities, grouped by category, and how each was classified for the production dataset.

4 Building the Provincial Production Dataset

The production dataset covers 42 of FABLE's 76 commodities across Punjab, Sindh, Khyber Pakhtunkhwa, Balochistan and a national total, for fiscal years 2000-01 through 2024-25. Every commodity follows the same six-column layout regardless of how complete its data is, and a missing year is always written as the literal string "no data" rather than estimated or left blank, so a gap looks like a gap and not like a zero.

Most crop figures come from Pakistan's Agricultural Statistics publications. A handful of commodities, including peas, sweet potato, coconut, and a few composite categories such as "Other fruits" and "Other vegetables," only appear in a separate horticultural survey published in five-year windows across five editions. Where a crop appears in both sources for overlapping years, the newer edition was used only to cross-check the existing figures rather than replace them, since later editions occasionally bundle several named crops into a single combined line for a given province, which can make a real crop look like it produced nothing that year if taken at face value. Where a crop genuinely only appears in one source, that source was used directly.

A few of FABLE's commodities are composites, built by summing several Pakistani crops together — rice combines two locally tracked varieties, for instance, and several "Other" categories sum minor crops that Pakistan reports individually but FABLE groups together. These groupings were fixed once at the start and applied consistently.

Three unit decisions are worth noting here. Cotton lint was reported in bales and converted to tonnes using Pakistan's standard bale weight. Eggs were reported as a count and converted to a weight basis using an average egg weight, since FABLE's own balance table works in tonnes for every commodity and a count cannot be joined against trade figures reported by weight. And a labelling error in three sheets had them recorded in raw tonnes instead of the workbook's standard unit of thousands of tonnes, which first became obvious as an implausible national total for one of the affected crops.

A set of Python scripts was built to standardise every sheet into this format: enforcing consistent units, checking that the four provincial figures sum to the national total for each year, and flagging any row where they do not. This makes the dataset something that can be regenerated and re-checked directly from its sources, rather than relying on one-off manual edits that would be hard to retrace later.

Figure 1 shows a finished sheet from the production dataset, including the automated province-sum check column described above.

#1 Wheat — Production ('000 Tonnes)

PBS ASP 2010-11 + MINFAL ASP 2021-22 + PBS District-Wise. Full detail -> Notes sheet.

Year	Punjab	Sindh	KPK	Balochistan	Pakistan	Province Sum Check
2000-01	15419.000	2226.500	764.000	614.200	19023.700	OK
2001-02	14594.400	2101.000	890.500	640.600	18226.500	OK
2002-03	15355.000	2109.200	1064.400	654.700	19183.300	OK
2003-04	15639.000	2172.200	1025.200	663.400	19499.800	OK
2004-05	17375.000	2508.600	1091.100	637.600	21612.300	OK
2005-06	16776.000	2750.300	1100.600	649.900	21276.800	OK
2006-07	17853.000	3409.200	1160.400	872.100	23294.700	OK
2007-08	15607.000	3411.400	1071.800	868.600	20958.800	OK
2008-09	18420.000	3540.200	1204.500	868.200	24032.900	OK
2009-10	17919.000	3703.100	1152.500	536.200	23310.800	OK
2010-11	19041.000	4287.900	1155.800	729.100	25213.800	OK
2011-12	17738.900	3761.500	1130.300	842.700	23473.400	OK
2012-13	18587.000	3598.700	1257.600	768.100	24211.400	OK
2013-14	19738.900	4002.100	1363.100	875.300	25979.400	OK
2014-15	19281.900	3672.200	1259.900	872.100	25086.100	OK
2015-16	19526.700	3834.600	1400.500	871.300	25633.100	OK
2016-17	20466.300	3910.400	1365.100	931.800	26673.600	OK
2017-18	19178.500	3639.500	1322.700	935.400	25076.100	OK
2018-19	18377.200	3778.900	1336.700	865.300	24358.100	OK
2019-20	19401.900	3848.100	1130.400	867.200	25247.600	OK
2020-21	20900.000	4043.200	1361.600	1159.300	27464.100	OK
2021-22	20032.000	3556.700	1362.800	1257.100	26208.700	OK
2022-23	21225.030	3940.170	1479.130	1516.230	28160.560	OK
2023-24	24243.300	4591.550	1493.800	1486.100	31814.740	OK
2024-25	22054.000	3542.510	1411.210	1470.570	28478.290	OK

Figure 1: A finished commodity sheet from the production dataset, with the automated province-sum check column on the right.

5 The Concordance Table

Every good that crosses Pakistan's border is recorded against a six-digit tariff code under the Harmonized System (HS), the same classification customs authorities everywhere use. FABLE, on the other hand, groups commodities by FAO item code, and the two classifications do not line up in any simple way — a single FABLE commodity can correspond to several HS codes, and a single HS code sometimes has to be split across more than one FABLE commodity, or left out altogether. Before any trade figures could be pulled, this translation between the two systems had to be worked out in full and written down in one place: the concordance table.

For every HS code that might plausibly belong to a FABLE commodity, the concordance records one of four actions. Some codes are included exactly as reported. Others are excluded entirely, either because they describe a manufactured product rather than a primary crop, because including them would double-count something already captured elsewhere, or because they belong to a different FABLE commodity. A third group needs a conversion factor before it can be added in at all, because the form Pakistan actually trades is not the form FABLE expects — milled rice rather than paddy, cocoa butter rather than cocoa beans, a live animal rather than a carcass. A final small group covers commodities Pakistan trades in genuinely negligible or zero quantity.

Conversion factors needed real sourcing rather than a rough guess, since the wrong factor distorts a commodity's entire trade figure. Where possible, a factor was taken from an authority specific to that commodity: FAO's own conversion tables for milling and livestock yields, the International Cocoa Organization for cocoa products, IRRI for rice, and industry references for smaller commodities such as guar gum, whose conversion factor went unsourced for a long stretch of the project before an appropriate industry reference was found. A small number of factors remain honest estimates rather than fully verified figures, and are labelled as such.

Figure 2 shows a section of the finished concordance table, with each HS code's action — include, exclude, or convert — colour-coded for quick reading.

FABLE-Pakistan Concordance Table -- Pakistani Crop -> FAO Item -> HS Code

530 mappings, 78 commodities | See the Notes sheet for column and Action definitions.

Pakistan_Crop_Name	FAO_Item_Name	FAO_Code	HS_Cod	HS_Description	FABLE_Category	Action	Conv_Facto
Wheat	Wheat	15	1001.10	Durum wheat (pre-2012 code)	CEREALS	INCLUDE	1.00
Wheat	Wheat	15	1001.90	Wheat other than durum (pre-2012 code)	CEREALS	INCLUDE	1.00
Wheat	Wheat	15	1001.11	Durum wheat, seed	CEREALS	INCLUDE	1.00
Wheat	Wheat	15	1001.19	Durum wheat, other	CEREALS	INCLUDE	1.00
Wheat	Wheat	15	1001.91	Common/spelt wheat, seed	CEREALS	INCLUDE	1.00
Wheat	Wheat	15	1001.99	Common/spelt wheat, other	CEREALS	INCLUDE	1.00
Wheat	Wheat and meslin flour	16	1101.00	Wheat flour	CEREALS	CONVERT	**1.03**
Basmati Rice + IRRI Rice	Rice, paddy	27	1006.10	Rice in husk (paddy/rough)	CEREALS	INCLUDE	1.00
Basmati Rice + IRRI Rice	Husked rice	28	1006.20	Husked (brown) rice	CEREALS	CONVERT	**1.25**
Basmati Rice + IRRI Rice	Rice; milled	31	1006.30	Semi-milled or wholly milled rice	CEREALS	CONVERT	**1.49**
Basmati Rice + IRRI Rice	Rice; broken	32	1006.40	Broken rice	CEREALS	CONVERT	**1.49**
Basmati Rice + IRRI Rice	Rice, paddy	27	1102.30	Rice flour	CEREALS	EXCLUDE	—
Maize	Maize	56	1005.10	Maize (corn) — seed	CEREALS	INCLUDE	1.00
Maize	Maize	56	1005.90	Maize (corn) — other	CEREALS	INCLUDE	1.00
Maize	Flour of maize	58	1102.20	Maize flour	CEREALS	CONVERT	**1.12**
Bajra	Millet	79	1008.20	Millet (pre-2012 code)	CEREALS	INCLUDE	1.00
Bajra	Millet	79	1008.21	Millet — seed	CEREALS	INCLUDE	1.00
Bajra	Millet	79	1008.29	Millet — other	CEREALS	INCLUDE	1.00
Jowar	Sorghum	83	1007.00	Grain sorghum (pre-2012 code)	CEREALS	INCLUDE	1.00
Jowar	Sorghum	83	1007.10	Grain sorghum — seed	CEREALS	INCLUDE	1.00
Jowar	Sorghum	83	1007.90	Grain sorghum — other	CEREALS	INCLUDE	1.00

Figure 2: A section of the finished concordance table, mapping Pakistani crop names through FAO item codes to individual HS tariff codes.

6 Building the Trade Dataset

Trade figures were pulled from UN Comtrade for every tariff code the concordance marks as includable or convertible, covering both import and export flows from 2000 through 2025. Pakistan filed no reporter submission at all for 2000 through 2002, so those three years were reconstructed instead from partner countries' own records — what every other country reported trading with Pakistan, with the flow direction reversed.

Figure 3 shows one commodity's finished trade sheet: value and quantity for both import and export flows, summed across every tariff code that maps to it.

Tobacco | FABLE group: BEVSPICES

3 HS code(s) summed. Qty = million tonnes; Value = US$.

Year	Export — Value (US$)	Import — Value (US$)	Export — Qty (M tonnes)	Import — Qty (M tonnes)
2000	2,912,596	2,719,969	0.001382	0.000980
2001	10,354,906	4,934,561	0.004020	0.002095
2002	6,242,871	2,838,863	0.002895	0.000810
2003	7,373,033	357,442	0.005463	0.000091
2004	10,569,087	1,059,308	0.007251	0.000299
2005	9,185,111	5,380,185	0.005760	0.002401
2006	7,026,931	7,287,899	0.004220	0.002220
2007	9,711,473	10,582,341	0.006323	0.002827
2008	7,031,476	9,234,185	0.002792	0.001845
2009	10,224,744	19,741,117	0.003937	0.002948
2010	16,132,508	12,609,726	0.005360	0.001672
2011	30,066,708	17,221,362	0.009569	0.001995
2012	23,181,074	12,377,956	0.007100	0.001518
2013	23,900,103	11,436,747	0.008059	0.001423
2014	21,432,757	15,713,791	0.006089	0.001965
2015	10,258,402	11,881,215	0.003224	0.001607
2016	9,896,689	11,724,063	0.002797	0.001814
2017	29,187,924	10,985,246	0.008015	0.001608
2018	20,789,245	14,341,463	0.007949	0.002026
2019	10,871,324	24,587,689	0.003868	0.003550
2020	24,285,652	9,255,240	0.009666	0.001806
2021	20,932,763	28,669,292	0.008252	0.009170
2022	40,750,291	29,933,827	0.014168	0.008868
2023	66,459,992	8,985,071	0.023660	0.004111
2024	107,824,612	5,954,394	0.028448	0.011443
2025	149,080,500	7,031,019	0.043069	0.008131

Figure 3: A finished commodity sheet from the aggregated trade dataset.

The most significant finding at this stage was that the Harmonized System itself gets revised periodically — in 2002, 2012, 2017 and 2022 across the years this project covers — and old tariff codes are retired and replaced at each revision. Comtrade answers a query using whichever version of the code system was in force that year, so a retired code returns normal-looking trade right up to the revision and then an exact zero afterward, which is indistinguishable from a real absence of trade unless it is specifically checked for. This is how Pakistan's kino (mandarin) exports, its single largest fruit export, appeared to collapse to almost nothing from 2017 onward: the tariff code it had been mapped to was split into three new codes that year, and the concordance was never updated to follow. The same check, applied systematically across every revision boundary rather than just this one case, turned up a much larger gap: the concordance had been built almost entirely from codes introduced by the 2012 revision, so an entire cluster of commodities — Wheat foremost among them — was running on codes that simply did not exist for 2003 through 2011, years that include two of FABLE's own calibration points. Restoring the missing codes recovered close to five billion dollars of trade across a dozen commodities.

A second, unrelated problem surfaced in Cotton lint, whose export total had been dominated for years by a tariff code describing cotton waste recovered during yarn spinning — a manufacturing byproduct, not something a cotton grower produces — which had inflated its official export figure several times over. Both problems are corrected in the finished trade dataset.

Figure 4 shows what the kino tariff-code fix did to the numbers.

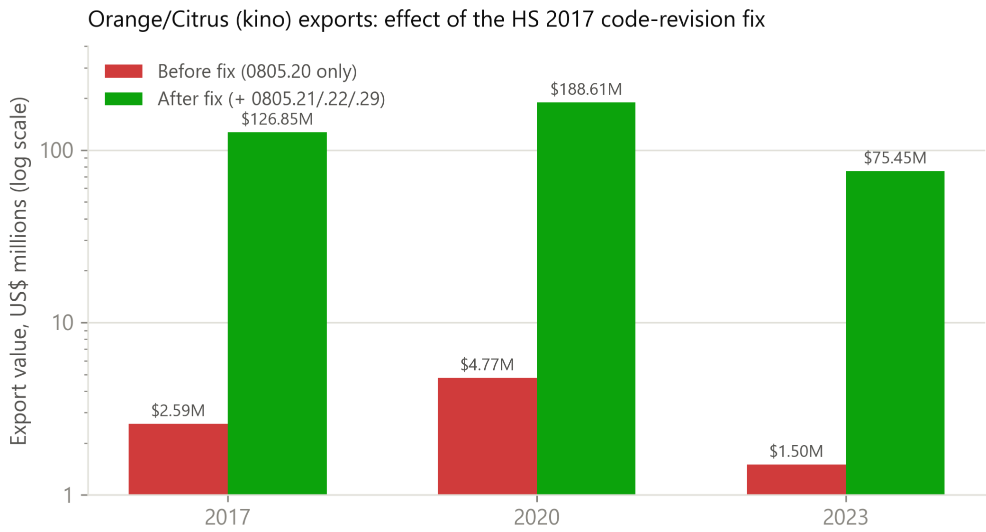

Figure 4: Orange/Citrus export value before and after adding the 2017 tariff-code successors for kino.

7 Joining Production and Trade

The finished production dataset and the finished trade dataset are joined into one national balance table — production plus imports minus exports — for every FABLE commodity, every year. Two things had to be resolved first. The two datasets identify commodities differently under the hood: production sheets are named after the commodity, while the trade side groups by tariff-code mapping, and joining on the name alone silently dropped five commodities whose names contain a forward slash, a character Excel does not allow in a sheet title. And production is reported by Pakistani fiscal year while trade is reported by calendar year, so each fiscal year was mapped to its first calendar year, matching the convention FAOSTAT itself uses for Pakistani crop years.

The balance is only calculated when production, imports and exports are all genuinely present for that commodity and year; a missing value is never assumed to be zero. The result is a complete 76-commodity by 26-year grid, so any gap in coverage shows up directly in the table.

Figure 5 shows this table in practice for Wheat.

1. Wheat						
FABLE group: CEREALS | Pakistani source: Wheat | Balance = Production + Imports − Exports (roughly, how much stayed in Pakistan that year).						
Year	Production (M tonnes)	Imports (M tonnes)	Exports (M tonnes)	Balance (M tonnes)	Import Value (US$)	Export Value (US$)
2000	19.023700	0.013027	0.003395	19.033333	3,739,966	474,090
2001	18.226500	0.005474	0.074638	18.157335	4,762,818	10,777,316
2002	19.183300	0.120989	0.638719	18.665570	14,364,930	91,740,359
2003	19.499800	0.000000	0.682650	18.817150	0	90,151,242
2004	21.612300	0.107978	0.343388	21.376890	23,252,508	48,648,035
2005	21.276800	0.772749	0.516316	21.533233	157,153,582	101,098,520
2006	23.294700	0.575884	0.584690	23.285894	107,426,614	119,704,846
2007	20.958800	0.192753	0.940933	20.210620	89,819,178	197,646,291
2008	24.032900	3.733868	0.174448	27.592320	1,628,717,842	50,389,561
2009	23.310800	1.106456	0.018101	24.399155	295,383,002	3,664,577
2010	25.213800	0.114195	0.160459	25.167536	64,172,014	39,309,768
2011	23.473400	0.047185	3.362927	20.157659	26,508,412	1,026,219,661
2012	24.211400	0.079644	1.084281	23.206763	25,091,138	293,517,974
2013	25.979400	0.410396	0.809867	25.579930	128,218,017	247,566,684
2014	25.086100	0.690134	0.474409	25.301825	190,020,345	203,506,741
2015	25.633100	0.012666	1.039372	24.606394	3,523,994	326,201,588
2016	26.673600	0.000123	0.613982	26.059742	37,713	174,514,925
2017	25.076100	0.000162	0.372131	24.704131	56,189	98,101,038
2018	24.358100	0.000994	2.159611	22.199483	269,530	442,055,803
2019	25.247600	0.000214	0.769146	24.478669	55,001	189,145,591
2020	27.464100	2.490642	0.002638	29.952104	660,087,296	807,058
2021	26.208700	2.486239	0.001205	28.693734	812,752,515	443,931
2022	28.160560	2.336609	0.001020	30.496149	943,432,383	340,006
2023	31.814740	2.605291	0.000053	34.419978	835,548,913	16,185
2024	28.478290	2.169334	0.000000	30.647624	636,224,554	0
2025	no data	0.000041	0.000000	no data	48	0

Figure 5: Wheat's finished sheet in the joined production–trade balance table.

The same series is plotted in Figure 6, including the spike in 2008 that reflects real wheat imports during that year's global food-price crisis — a spike that only appears because of the tariff-code fix described in the previous section; the original, unrevised trade pull had missed almost all of it.

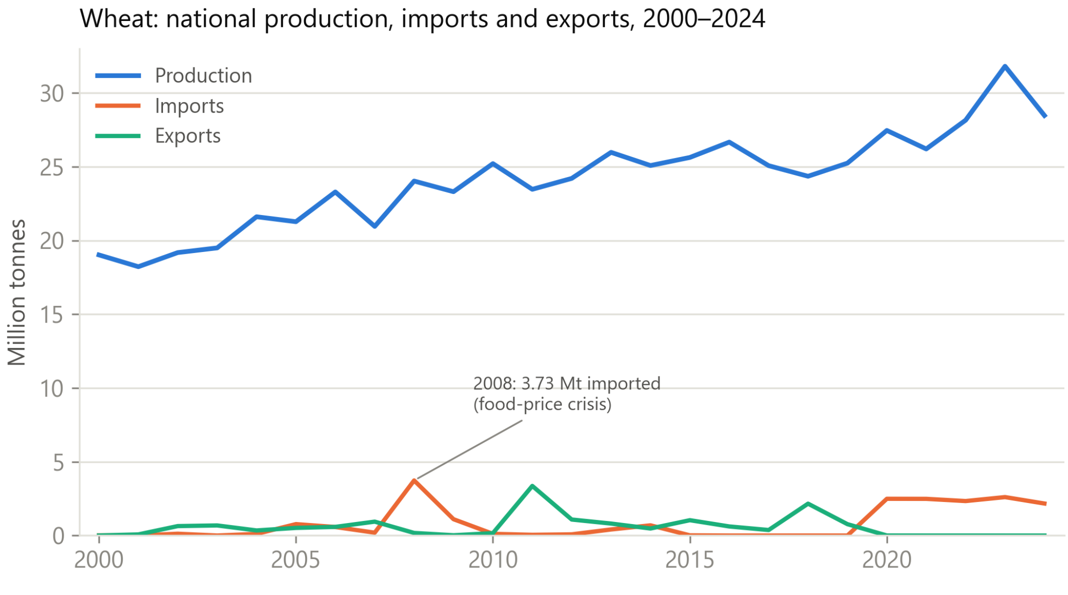

Figure 6: Wheat's national production, import and export series in the final joined dataset.

8 Validating Against FAOSTAT

The last stage compares the project's trade figures against FAOSTAT's published Pakistan trade data, commodity by commodity and year by year, across export and import value and quantity — 7,904 comparable cells in total.

Figure 7 shows one commodity's finished validation sheet, with each percentage difference colour-coded and a short written note explaining the pattern for that commodity.

#1 Wheat (CEREALS)

Coverage: FULL. Green <= 10% diff / red > 10% / grey = no comparison. See the Contents sheet for coverage and notes.

Remarks: Good agreement from 2011 onward; nearly all the red is in 2000-2010. For 2000-2002 our figures use partner-country mirror data (Pakistan did not report to Comtrade those years, so trade is under-captured); for 2003-2010 FAOSTAT filled much of Pakistan's grain trade with its own estimates. Wheat is also dominated by government-to-government import deals and strategic reserves, which the two sources record and value on different timings.

Year	Our Export Value (US$)	FAO Export Value (US$)	Export Value Diff %	Our Export Qty (M tonnes)	FAO Export Qty (M tonnes)	Export Qty Diff %	Our Import Value (US$)	FAO Import Value (US$)	Import Value Diff %	Our Import Qty (M tonnes)	FAO Import Qty (M tonnes)	Import Qty Diff %
2000	474089.51	9667000	-95.1%	0.003394964	0.05292221	-93.6%	3739966	144207000	-97.4%	0.01302749	1.048179	-98.8%
2001	10777315.88	81421000	-86.8%	0.074638391	0.77162044	-90.3%	4762818	26725000	-82.2%	0.005473622	0.149121	-96.3%
2002	91740358.77	130806000	-29.9%	0.6387186	1.19667732	-46.6%	14364930	51154000	-71.9%	0.120988744	0.26721466	-54.7%
2003	90151242	185729000	-51.5%	0.68264998	1.54885342	-55.9%	0	29658000	-100.0%	0	0.14895021	-100.0%
2004	48648035	48648000	0.0%	0.343388168	0.34338816	0.0%	23252508	23264000	0.0%	0.107978	0.107978	0.0%
2005	101098520	95641000	5.7%	0.51631634	0.71656714	-27.9%	157153582	299137000	-47.5%	0.77274886	1.44049936	-46.4%
2006	119704846	97216000	23.1%	0.584689833	0.48646488	20.2%	107426614	149464000	-28.1%	0.575884038	0.8626995	-33.2%
2007	197646291	206951000	-4.5%	0.940933338	0.99219746	-5.2%	89819178	41450000	116.7%	0.19275315	0.13599708	41.7%
2008	50389561	55538000	-9.3%	0.17444754	0.2934328	-40.5%	1628717842	782467000	108.2%	3.733867938	1.82984108	104.1%
2009	3664577	37512000	-90.2%	0.0181009	0.142506	-87.3%	295383002	1040473000	-71.6%	1.106456368	3.10265975	-64.3%
2010	39309768	2146000	1731.8%	0.16045949	0.01028437	1460.2%	64172014	86832000	-26.1%	0.11419546	0.17458031	-34.6%
2011	1026219661	1026438000	0.0%	3.362926611	3.36369706	0.0%	26508412	26517000	0.0%	0.047185184	0.04718518	0.0%
2012	293517974	293304000	0.1%	1.084280949	1.08532525	-0.1%	25091138	27290000	-8.1%	0.07964372	0.08357111	-4.7%
2013	247566684	248477000	-0.4%	0.809866675	0.81134676	-0.2%	128218017	130009000	-1.4%	0.410396446	0.41356265	-0.8%
2014	203506741	203805000	-0.1%	0.474408937	0.474796286	-0.1%	190020345	190030000	0.0%	0.690134108	0.69015059	0.0%
2015	326201588	326324000	0.0%	1.039371958	1.03964388	0.0%	3523994	3544000	-0.6%	0.01266615	0.01269602	-0.2%
2016	174514925	174564000	0.0%	0.613981672	0.61408983	0.0%	37713	40000	-5.7%	0.0001234	0.000124904	-1.2%
2017	98101038.09	98102000	0.0%	0.37213106	0.37118114	0.3%	56188.5	58000	-3.1%	0.00016168	0.000136757	18.2%
2018	442055803.5	442146000	0.0%	2.15961104	2.15986545	0.0%	269530.2	270000	-0.2%	0.00099395	0.00099395	0.0%
2019	189145591.2	189555000	-0.2%	0.76914557	0.77093468	-0.2%	55001.1	55000	0.0%	0.00021424	0.00021424	0.0%
2020	807057.86	6339000	-87.3%	0.00263783	0.03063783	-91.4%	660087296.1	660095000	0.0%	2.490642139	2.490670333	0.0%
2021	443930.64	444000	0.0%	0.0012051	0.0012051	0.0%	812752514.6	812752000	0.0%	2.48623941	2.48623941	0.0%
2022	340005.97	340000	0.0%	0.0010197	0.0010197	0.0%	943432382.7	943432000	0.0%	2.336609	2.33682839	0.0%
2023	16184.52	16000	1.2%	0.00005253	0.00005253	0.0%	835548913.5	835549000	0.0%	2.605291	2.605291	0.0%
2024	0	no data	no data	0	no data	no data	636224554.4	636225000	0.0%	2.169334	2.169334	0.0%
2025	0	no data	no data	0	no data	no data	47.99	no data	no data	0.000041	no data	no data

Figure 7: Wheat's finished validation sheet, comparing the project's trade figures against FAOSTAT year by year.

Building this comparison also helped catch mistakes made earlier in the pipeline. Several commodities that looked badly mismatched against FAOSTAT turned out, on closer inspection, to be mapped incorrectly on this project's own side rather than genuinely different from FAOSTAT's figures — a reference code pointing at the wrong FAOSTAT item in one case, a comparison against the wrong form of a converted product in another. Correcting these moved several major commodities from mostly disagreeing with FAOSTAT to matching it closely.

After every correction, a majority of comparable cells land within ten percent of FAOSTAT's own figure. That share is not spread evenly across the years, though, and understanding why turned out to matter. Agreement is low and inconsistent through 2010, then settles into a stable, much higher range from 2011 onward, as Figure 8 shows.

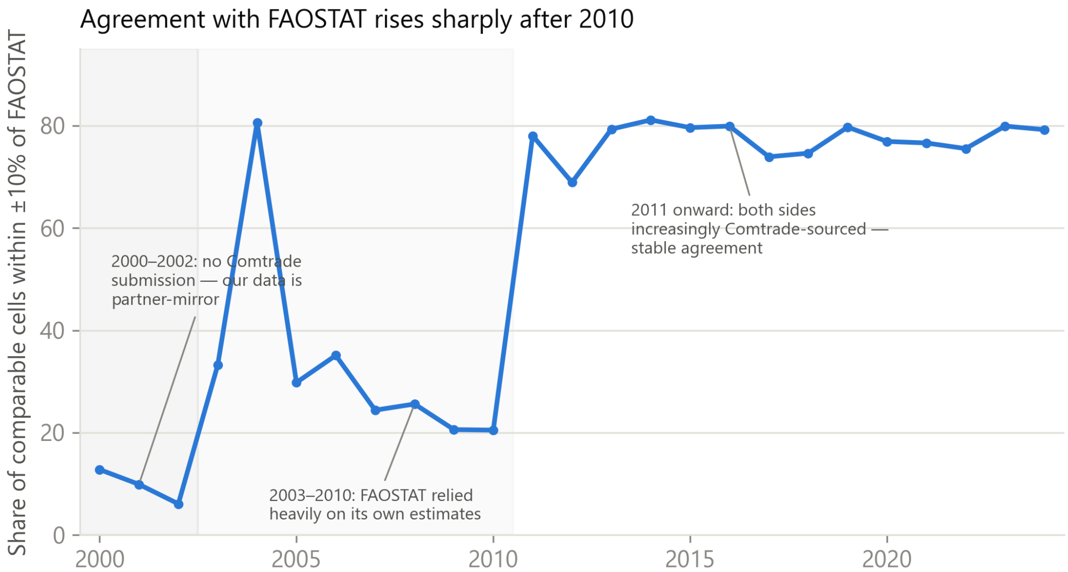

Figure 8: Share of comparable cells matching FAOSTAT within ±10%, by year.

The 2000–2002 years reflect a genuine limitation on this project's own side: Pakistan filed nothing with Comtrade in those years, so the figures for that period had to be reconstructed from partner-country records, which tend to undercount. The 2003–2010 years are closer to the reverse situation — FAOSTAT's own published data-quality indicators show that a large share of Pakistan's trade rows in that window were estimated rather than drawn from actual customs records, a share that falls close to zero from 2014 onward. From 2011 on, both sources increasingly draw from the same underlying Comtrade data, which is why they converge. This points to a conclusion worth stating plainly: the early-year gap should not be closed by adjusting this project's own figures to match FAOSTAT's for that period, since those figures are the less reliable side of the comparison for that stretch of years, not the other way round.

A smaller set of mismatches reflect a genuine difference in scope rather than an error on either side — one vegetable category, for example, includes garlic, which Pakistan imports in real volume but which FAOSTAT tracks under no item at all, so that comparison will always run high on one side. Cases like this are documented individually in the validation output.

9 Limitations

This dataset is only as complete as Pakistan's own published statistics allow. Livestock products — milk, beef, mutton, poultry, eggs and animal fat — have no provincial breakdown in any source checked, so every livestock figure here is a national total. A handful of year ranges remain unfillable with the sources currently available, most notably livestock data for 2011-12 through 2017-18, where the available editions simply do not print those years. A few conversion factors, including one for rice flour and one catch-all milk-powder category, are honest estimates rather than fully verified figures, and are labelled as such. A small number of real Pakistani trade flows, including pepper, sit outside FABLE's own commodity classification entirely and would need a decision on where they belong before being added. And, as noted in Table 2, three commodities —

cotton seed, butter and ghee, and one citrus sub-category — were initially expected to have usable production data, but turned out, once the actual publications were checked, to have no separate series after all.

10 Conclusion

This project set out to give FABLE a Pakistan-specific alternative to its default FAOSTAT inputs, and to understand where and why the two would disagree once that alternative existed. Both parts of that goal were met: a production dataset and a trade dataset covering most of FABLE's 76 commodities across 25 years, joined into one balance table and checked line by line against FAOSTAT. The more lasting lesson from the process, though, is how many of the biggest findings came down to simply not accepting a plausible-looking number without checking it first — a zero that turned out to be a retired tariff code, a total dominated by a byproduct rather than the product it claimed to be, a join that silently dropped a fifth of its matches. None of these looked wrong from the outside. That habit of double-checking, more than any single figure in this dataset, is what carries forward into whatever comes next.

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